

Olympic Medal Outcome Prediction System

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Phase 2 Report

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1. Problem Description

The objective of this phase is to develop and analyze machine learning models for predicting the medal outcome of athletes using historical Olympic data. The dataset (`athlete_events.csv`) contains 271,116 records across 15 features including Age, Height, Weight, Sex, Team, Sport, and Medal.

In this phase, two machine learning algorithms — Logistic Regression and K-Nearest Neighbors (KNN) — are implemented and trained on the dataset. Their performance is evaluated using accuracy score. A preliminary comparative analysis is performed to understand which model shows better promise for medal outcome classification. Further models and deeper evaluation are planned for Phase 3.

Key aspects of the problem:

- Dataset Size: 271,116 athlete records spanning Olympics from 1896 to 2016.
- Target Variable: Medal outcome — Gold (13,372), Bronze (13,295), Silver (13,116).
- Input Features: Sex, Age, Height, Weight, Team, Sport (after preprocessing).
- Task Type: Multi-class classification — predicting one of 3 medal types.
- Challenge: Handling missing values in Height, Weight, and Medal columns.

2. Architecture

The system follows a standard supervised machine learning pipeline as described below:



Fig. 1 — System Architecture Flow Diagram (Phase 2)

3. Models Used (Applied / Planned)

In Phase 2, two machine learning models have been applied. A third model (Linear Regression) is planned for Phase 3 for comparative analysis.

3.1 Logistic Regression (Applied)

Logistic Regression is a supervised classification algorithm that models the probability of a categorical outcome using the logistic (sigmoid) function. It is well-suited for multi-class classification problems.

- Configuration: max_iter=1000 to ensure convergence on large dataset.
- Features used: Sex, Age, Height, Weight, Team, Sport (all label encoded).
- Preliminary Accuracy: 85.38%
- Observation: Shows strong performance as a baseline model.

3.2 K-Nearest Neighbors — KNN (Applied)

KNN is an instance-based, non-parametric classification algorithm. It classifies a new data point based on the majority class among its K nearest neighbors in feature space. No explicit training phase is required.

- Configuration: n_neighbors=5 for initial run.
- Preliminary Accuracy: 83.91%
- Observation: Performs well but slightly below Logistic Regression at K=5.
- Note: Hyperparameter tuning (K vs Accuracy plot) is planned for Phase 3.

3.3 Linear Regression (Planned for Phase 3)

Linear Regression will be applied in Phase 3 for comparative purposes. It models the relationship between input variables and a continuous output. Although not ideally suited for classification, it will be evaluated using MSE and R^2 score to demonstrate the contrast with classification-specific models.

4. Evaluation Metrics Used

In Phase 2, the following preliminary evaluation metric has been used:

4.1 Accuracy Score (Applied)

Accuracy measures the proportion of correctly predicted instances out of the total number of instances. It is the primary metric used in this phase for a quick comparison between the two models.

Formula: $\text{Accuracy} = (\text{Number of Correct Predictions}) / (\text{Total Predictions})$

Model	Accuracy Score
Logistic Regression	85.38%
KNN (K=5)	83.91%

4.2 Metrics Planned for Phase 3

- Precision, Recall, F1-Score — via Classification Report
- Confusion Matrix — to visualize per-class prediction performance
- MSE and R² Score — for Linear Regression evaluation
- KNN Accuracy vs K plot — for hyperparameter tuning
- Final Model Comparison Bar Chart — to compare all three models

5. Preliminary Results and Visualizations

5.1 Dataset Overview

The Olympic dataset (athlete_events.csv) loaded and explored with the following key observations:

- Shape: 271,116 rows × 15 columns.
- Columns: ID, Name, Sex, Age, Height, Weight, Team, NOC, Games, Year, Season, City, Sport, Event, Medal.
- Medal value counts (before dropping NaN): Gold — 13,372 | Bronze — 13,295 | Silver — 13,116.
- Missing values in Height, Weight handled using column mean; Medal NaN filled with 'No Medal'.
- Final working columns selected: Sex, Age, Height, Weight, Team, Sport, Medal.

5.2 Total Medals Over Years (Line Plot)

A line plot was generated showing total medals awarded per Olympic year. The trend shows a consistent increase in total medals from the early 1900s, with significant growth post-1960 due to expansion of events and sports categories.

5.3 Preliminary Accuracy Comparison

Based on initial training and testing, Logistic Regression achieved 85.38% accuracy while KNN achieved 83.91% at K=5. Logistic Regression appears to be the stronger baseline model. A detailed comparison including Linear Regression and advanced visualizations will be conducted in Phase 3.

6. References

Books:

1. Introduction to Machine Learning with Python – Andreas C. Muller & Sarah Guido
2. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow – Aurelien Geron

Dataset:

- Olympic Athletes Dataset (athlete_events.csv) — Kaggle:
<https://www.kaggle.com/datasets/heesoo37/120-years-of-olympic-history-athletes-and-results>

Online Resources:

- Scikit-learn Documentation — <https://scikit-learn.org/stable/>
- GeeksforGeeks — <https://www.geeksforgeeks.org/>
- Stack Overflow — <https://stackoverflow.com>
- Matplotlib Documentation — <https://matplotlib.org/>